

Erasmus Mundus Master Programme EU-CORE
University of Zagreb Faculty of Electrical Engineering and Computing (FER)

Optimal EV Charging Scheduling for a Solar-Powered Parking Lot Using PV and Battery Storage

Student Project Control, Estimation and Optimization in Solar Energy

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Abstract

The increasing penetration of electric vehicles (EVs) introduces significant challenges for local energy systems due to the high and stochastic charging demand. Uncoordinated charging strategies may increase peak demand, operational costs and grid stress while reducing the utilization of locally available renewable generation. This work investigates the implementation of predictive smart charging in a solar-powered parking lot equipped with photovoltaic (PV) generation and battery energy storage systems (BESS).

Two charging strategies are compared: (i) uncontrolled as-soon-as-possible (ASAP) charging and (ii) controllable smart charging based on day-ahead optimization. The developed framework integrates yearly EV charging demand, fixed building consumption, photovoltaic production and electricity market prices. Representative low-, medium- and high-demand days are extracted from the yearly dataset and analyzed.

Results demonstrate that smart charging reduces daily operating costs by up to 15.4

1 Introduction

Electrification of transportation constitutes a key component of the global decarbonization strategy. However, the increasing deployment of electric vehicles imposes additional challenges on distribution grids due to concentrated charging demand.

Conventional uncontrolled charging, commonly referred to as *as-soon-as-possible* charging, initiates charging immediately after vehicle arrival. Although simple to implement, this strategy frequently leads to:

- increased peak demand,
- higher electricity procurement costs,
- poor utilization of renewable generation,
- increased stress on distribution infrastructure.

Smart charging represents an attractive alternative because charging power can be shifted within the available parking interval while still satisfying user energy requirements.

The objective of this project is therefore to quantify the operational and economic benefits achieved by introducing controllable EV charging in a solar-powered parking lot equipped with PV generation and battery energy storage.

2 Data Description

2.1 EV Charging Dataset

EV charging sessions were synthetically generated using an EV generative model. The generation procedure considered realistic daily vehicle arrival patterns.

The yearly dataset contains:

- 365 simulated days;
- 8,530 charging sessions;
- average daily number of sessions: 23.37;
- annual EV charging demand: 95,663 kWh.

Weekdays contain between 25 and 35 charging sessions, whereas weekends contain between 5 and 10 sessions.

2.2 Fixed Building Load

The non-EV load profile originates from the LargeHotel dataset. The original normalized profile was scaled such that the total annual building energy consumption equals the total annual EV charging demand:

$$E_{load,annual} = E_{EV,annual} = 95,663 \text{ kWh} \quad (1)$$

This assumption enables balanced analysis of EV demand impact.

2.3 Photovoltaic Production and Electricity Prices

Hourly normalized PV production and hourly electricity market prices were employed.

The available photovoltaic power is computed as

$$P_{PV}(t) = \zeta_{PV}(t)S_{PV} \quad (2)$$

where:

- $\zeta_{PV}(t)$ denotes normalized PV production;
- S_{PV} denotes installed PV capacity.

3 Methodology

3.1 Baseline Charging Strategy

In the baseline scenario, EV charging is considered non-controllable.

Each charging session starts immediately after vehicle connection and continues until the required energy is delivered.

Consequently,

$$P_{load}^{tot}(t) = P_{fixed}(t) + P_{EV}^{ASAP}(t) \quad (3)$$

where:

- $P_{fixed}(t)$ denotes fixed building consumption;
- $P_{EV}^{ASAP}(t)$ denotes uncontrolled EV charging demand.

3.2 Smart Charging Strategy

For smart charging, EV charging becomes a decision variable.

The optimization determines the charging profile while satisfying

$$\sum_{t=t_{arr}}^{t_{full}} P_{EV}(t) \Delta t = E_{req} \quad (4)$$

subject to

$$0 \leq P_{EV}(t) \leq P_{max} \quad (5)$$

The optimizer minimizes total operating costs while respecting charging deadlines.

3.3 Objective Function

The optimization minimizes

$$J = J_E + J_{pp} + J_{deg} \quad (6)$$

where:

- J_E : electricity purchase cost,
- J_{pp} : peak power penalty,
- J_{deg} : battery degradation cost.

4 Representative Day Selection

Three representative days were extracted:

- Low-demand day (Day 183),
- Average-demand day (Day 23),
- High-demand day (Day 153).

These days represent different charging demand conditions and enable systematic comparison.

5 Results

5.1 Economic Comparison

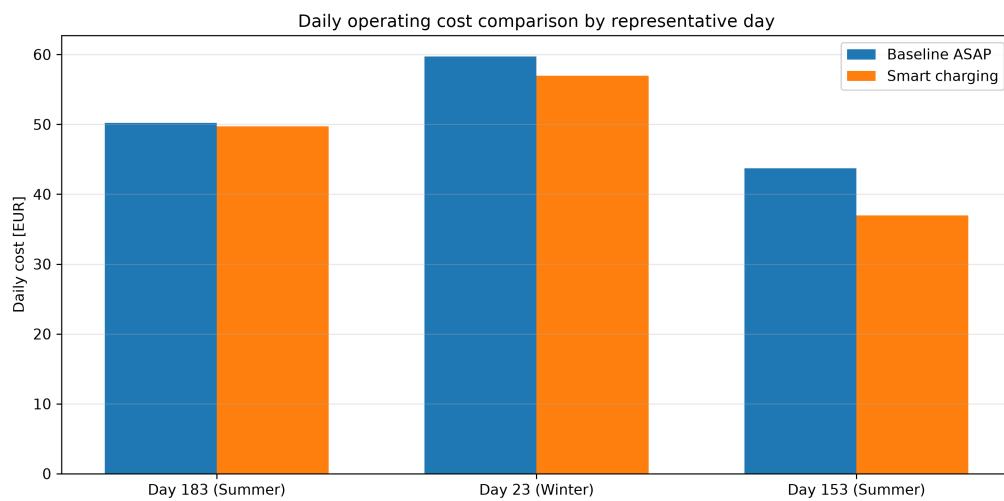


Figure 1: Comparison between baseline and smart charging daily operating costs.

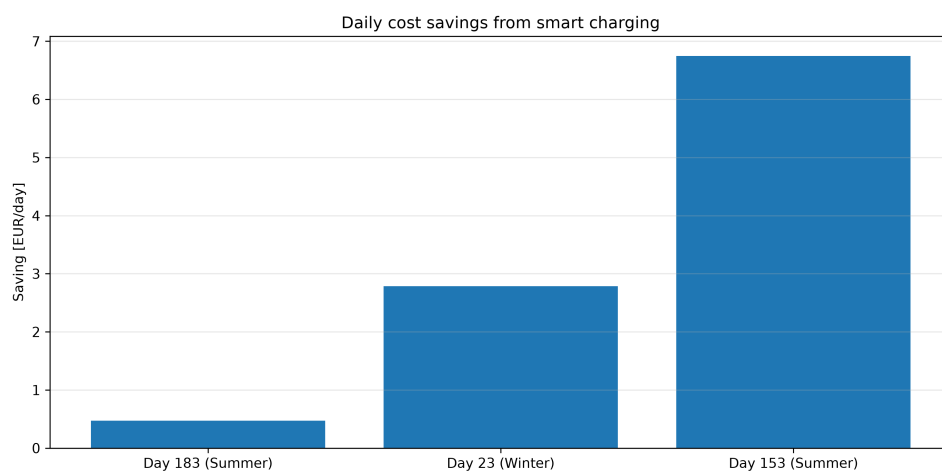


Figure 2: Daily cost savings achieved through smart charging.

5.2 Peak Demand Reduction

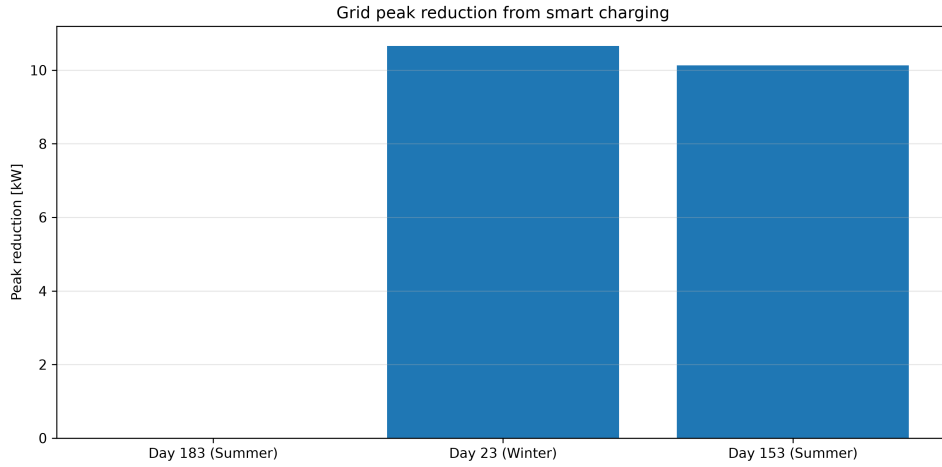


Figure 3: Grid peak demand reduction obtained using smart charging.

5.3 Detailed Daily Analysis

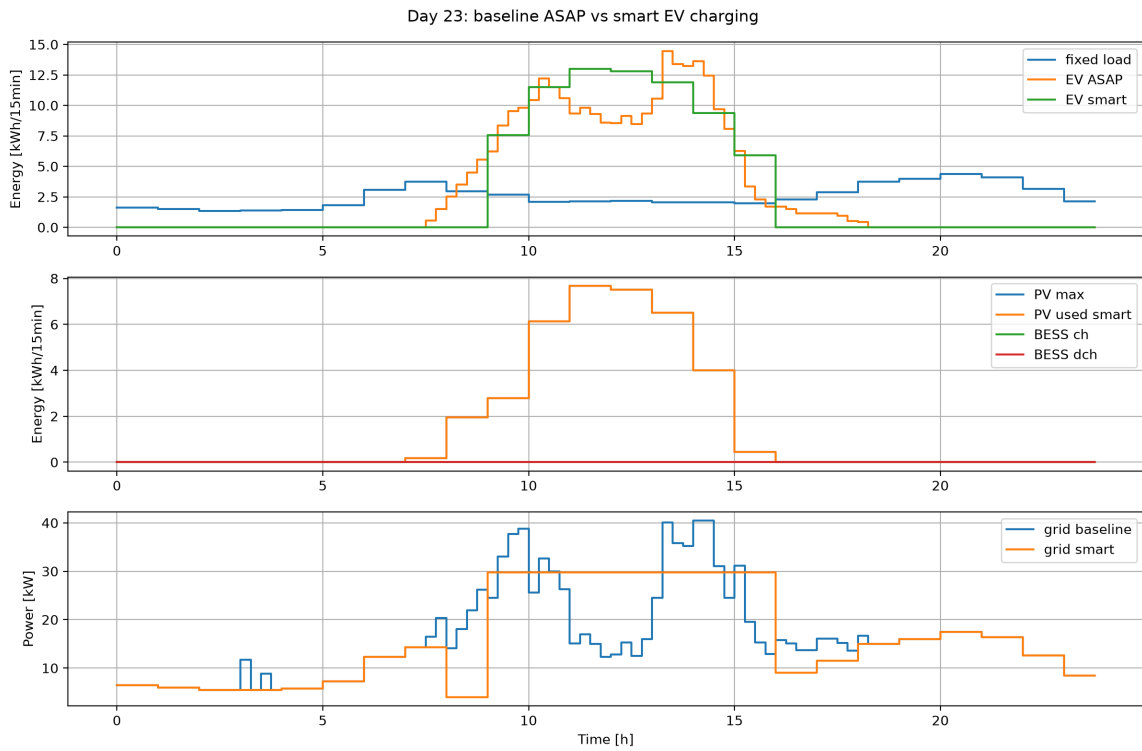


Figure 4: Detailed comparison for the representative average-demand day.

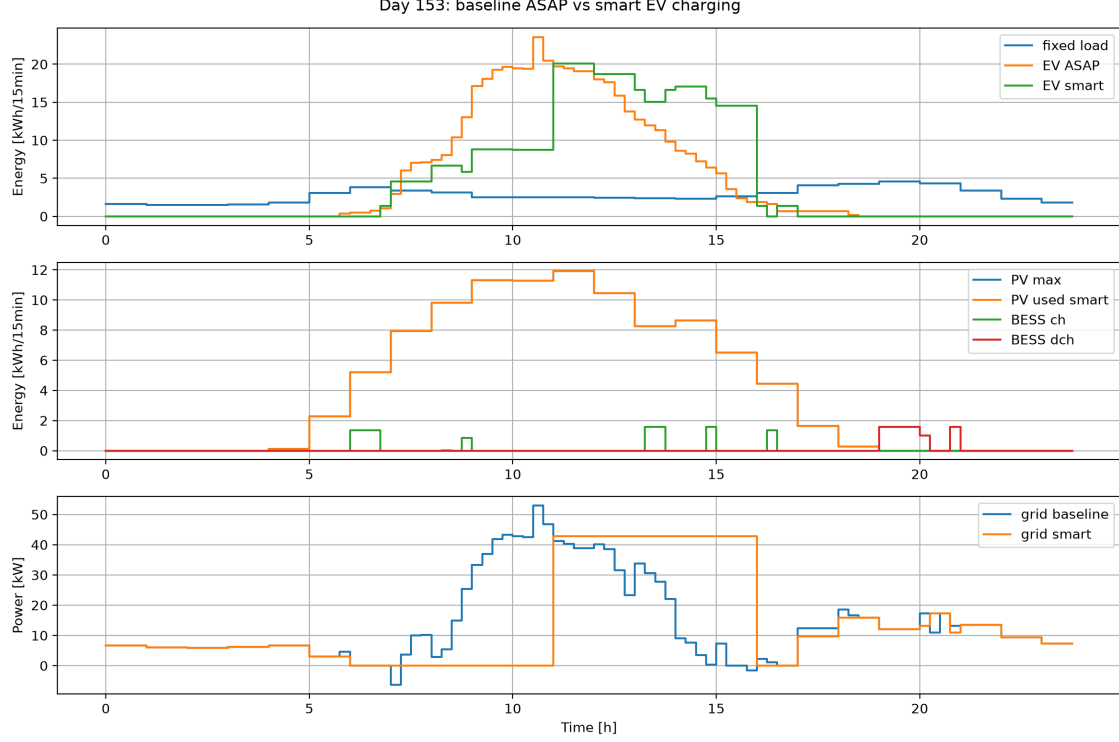


Figure 5: Detailed comparison for the representative high-demand day.

Table 1: Performance comparison between baseline and smart charging strategies.

Representative Day	Savings [EUR]	Savings [%]	Peak Reduction [kW]
Day 183 (Summer)	0.47	0.94	0.00
Day 23 (Winter)	2.79	4.67	10.66
Day 153 (Summer)	6.75	15.44	10.13

Results indicate that smart charging effectiveness strongly depends on charging demand.

For low-demand days only marginal improvements are observed because limited flexibility exists.

Conversely, medium- and high-demand days exhibit significant cost reductions and substantial peak shaving.

6 Discussion

The developed optimization framework successfully shifts EV charging towards periods with lower electricity prices and increased photovoltaic availability.

The results clearly demonstrate that:

1. Smart charging decreases operational costs.
2. Smart charging significantly reduces peak demand.

3. Benefits increase with higher EV penetration.

 Local renewable generation utilization improves.

Therefore, predictive EV charging constitutes an effective energy management strategy for solar-powered parking lots.

7 Conclusion

This work presented a day-ahead smart charging framework for electric vehicle charging in a solar-powered parking lot equipped with photovoltaic generation and battery energy storage.

Two charging strategies were investigated: uncontrolled as-soon-as-possible (ASAP) charging and controllable smart charging based on optimization.

The obtained results clearly demonstrate the advantages of predictive EV charging. For the representative low-demand day, only marginal improvements were observed, resulting in cost savings of approximately 0.9%. However, for medium- and high-demand scenarios, significantly larger benefits were achieved.

In particular, smart charging reduced daily operating costs by up to 15.4%, corresponding to savings of approximately 6.75 EUR/day. Furthermore, the optimization strategy reduced the maximum grid power demand by more than 10 kW for both medium- and high-demand days. Such peak shaving capabilities are particularly important for reducing stress on distribution infrastructure and avoiding costly grid reinforcement.

The results also showed that the economic benefits of smart charging increase with EV penetration. Higher charging demand provides greater flexibility for the optimization algorithm to shift charging towards periods with lower electricity prices and increased photovoltaic availability.

Overall, the developed framework successfully demonstrated that predictive smart charging improves the operational and economic performance of solar-powered parking facilities. The proposed methodology therefore represents a promising approach for future smart grid and renewable energy applications.

Future work may consider stochastic EV arrivals, uncertainty-aware optimization, vehicle-to-grid (V2G) operation and real-time model predictive control implementations.